

FORGE-X Investigation Report

DEMO-Corp-Breach-2026 — full report

Target	workstation-07.demo.lab (WINDOWS)	Source mode	DATASET
Provenance	SYNTHETIC	Generated	2026-09-26T18:40:03.199891+00:00
Report type	FULL	Author	lead
Evidence items	33	Findings	13

NOTE: This case was generated from SYNTHETIC sample data for demonstration. It is not real-world evidence.

1. Evidence Summary

Type	Count
FILE	3
NETWORK_CONNECTION	10
PROCESS	4
SYSTEM_EVENT	13
USER_ACCOUNT	3

Evidence integrity (SHA-256 anchor chain): VALID

33 anchor records · SHA-256 append-only ledger (full per-record recompute)

2. Findings

Severity	Title	Source	Status	Evidence refs
HIGH	Encoded command executed by powershell.exe (PID 4567)	SCRIPT	OPEN	a71890be
CRITICAL	Office application spawned powershell.exe	SCRIPT	OPEN	a71890be
HIGH	Brute-force pattern: 8 failed logins for user 'svc_backup'	SCRIPT	OPEN	62c19757, 6950aceb, c6756a7e, f2475860
CRITICAL	Account 'svc_backup' succeeded after 8 failed logins	SCRIPT	OPEN	62c19757, 6950aceb, c6756a7e, f2475860
HIGH	Persistence event: Scheduled task 'UpdaterSvc' created pointing to C:\Temp\updater.exe	SCRIPT	OPEN	508cdec9
HIGH	Connection to uncommon port 4444 (203.0.113.9)	SCRIPT	OPEN	37dc5a94
HIGH	Large outbound transfer: 5242880 bytes to 203.0.113.9:4444	SCRIPT	OPEN	37dc5a94
MEDIUM	Suspicious DNS lookup: cdn-update.ngrok.io	SCRIPT	OPEN	44b41760
HIGH	Connection to uncommon port 4444 (203.0.113.9)	SCRIPT	OPEN	6537944c
HIGH	Large outbound transfer: 5242880 bytes to 203.0.113.9:4444	SCRIPT	OPEN	6537944c
HIGH	Connection to uncommon port 4444 (203.0.113.9)	SCRIPT	OPEN	adedb791

Severity	Title	Source	Status	Evidence refs
HIGH	Connection to uncommon port 4444 (203.0.113.9)	SCRIPT	OPEN	cf664d5e
MEDIUM	Suspicious DNS lookup: cdn-update.ngrok.io	SCRIPT	OPEN	02505bcb

3. AI Incident Reconstruction

Engine: HEURISTIC · Model: forgex-correlation-engine-v1 · Confidence: 0.9

What happened: Correlated 33 evidence items into 6 attack stages (Initial Access → Execution → Persistence → Command and Control → Exfiltration → Lateral Movement). [Initial Access] Repeated authentication failures against account 'svc_backup' consistent with password guessing. [Execution] Obfuscated command line executed by powershell.exe. [Persistence] Scheduled task 'UpdaterSvc' created pointing to C:\Temp\updater.exe [Command and Control] Outbound connection to 203.0.113.9:4444 on a port commonly used by remote access tooling.

- **Initial Access:** Repeated authentication failures against account 'svc_backup' consistent with password guessing. (evidence: 484abd57, 5a195e54, 62c19757, 6950aceb, 7e660e00, 999c018d, a71890be, c6756a7e, dcdaa736, f2475860)
- **Execution:** Obfuscated command line executed by powershell.exe. (evidence: 16a587a7, 439024ca, a71890be)
- **Persistence:** Scheduled task 'UpdaterSvc' created pointing to C:\Temp\updater.exe (evidence: 508cdec9)
- **Command and Control:** Outbound connection to 203.0.113.9:4444 on a port commonly used by remote access tooling. (evidence: 02505bcb, 37dc5a94, 44b41760, 6537944c, adedb791, cf664d5e)
- **Exfiltration:** High-volume outbound transfer to 203.0.113.9 consistent with data exfiltration. (evidence: 37dc5a94, 6537944c)
- **Lateral Movement:** Connection to remote administrative service on 198.51.100.23. (evidence: c430d06e)

MITRE ATT&CK mapping: T1110 (Brute Force), T1078 (Valid Accounts), T1059.001 (PowerShell), T1204.002 (User Execution: Malicious File), T1105 (Ingress Tool Transfer), T1204.001 (User Execution: Malicious Link), T1571 (Non-Standard Port), T1048 (Exfiltration Over Alternative Protocol), T1021 (Remote Services), T1071.004 (Application Layer Protocol: DNS), T1547.001 (Registry Run Keys / Startup Folder)

Analysis gaps: No memory artifacts — in-memory activity cannot be confirmed.

4. Investigation Timeline (first 120 entries)

Timestamp	Type	Severity	Description
2026-09-26T10:00:20	PROCESS	INFO	outlook.exe (PID 2200) spawned by explorer.exe (svc_backup)
2026-09-26T10:00:30	FILE	INFO	C:\Users\svc_backup\Documents\Q3-financials.xlsx written/observed (1048576 bytes)
2026-09-26T10:02:12	SYSTEM_EVENT	MEDIUM	[login_failure] Failed password for svc_backup from 198.51.100.23 port 5000 ssh2
2026-09-26T10:03:12	SYSTEM_EVENT	MEDIUM	[login_failure] Failed password for svc_backup from 198.51.100.23 port 5010 ssh2
2026-09-26T10:04:12	SYSTEM_EVENT	MEDIUM	[login_failure] Failed password for svc_backup from 198.51.100.23 port 5020 ssh2
2026-09-26T10:05:12	SYSTEM_EVENT	MEDIUM	[login_failure] Failed password for svc_backup from 198.51.100.23 port 5030 ssh2
2026-09-26T10:06:12	SYSTEM_EVENT	MEDIUM	[login_failure] Failed password for svc_backup from 198.51.100.23 port 5040 ssh2
2026-09-26T10:07:12	SYSTEM_EVENT	MEDIUM	[login_failure] Failed password for svc_backup from 198.51.100.23 port 5050 ssh2
2026-09-26T10:08:12	SYSTEM_EVENT	MEDIUM	[login_failure] Failed password for svc_backup from 198.51.100.23 port 5060 ssh2
2026-09-26T10:09:12	SYSTEM_EVENT	MEDIUM	[login_failure] Failed password for svc_backup from 198.51.100.23 port 5070 ssh2
2026-09-26T10:12:44	SYSTEM_EVENT	LOW	[login_success] Accepted password for svc_backup from 198.51.100.23 port 5911 ssh2
2026-09-26T10:13:30	SYSTEM_EVENT	HIGH	[privilege_use] sudo: svc_backup : command=/usr/bin/install -m 755 updater.exe /tmp/updater.exe
2026-09-26T10:13:40	FILE	MEDIUM	C:\Temp\updater.exe written/observed (482304 bytes)
2026-09-26T10:14:10	PROCESS	HIGH	powershell.exe (PID 4567) spawned by outlook.exe (svc_backup)
2026-09-26T10:14:10	SYSTEM_EVENT	INFO	[process_create] Process created: powershell.exe -enc JABwAHIAbwBjACAAPQAg...
2026-09-26T10:15:00	PROCESS	INFO	updater.exe (PID 4600) spawned by powershell.exe (svc_backup)
2026-09-26T10:15:02	SYSTEM_EVENT	HIGH	[persistence] Scheduled task 'UpdaterSvc' created pointing to C:\Temp\updater.exe
2026-09-26T10:16:10	FILE	INFO	C:\Temp\data.zip written/observed (5242880 bytes)
2026-09-26T10:16:15	PROCESS	HIGH	curl.exe (PID 4700) spawned by updater.exe (svc_backup)
2026-09-26T10:16:20	SYSTEM_EVENT	INFO	[network_connect] Windows Filter Platform permitted connection to 203.0.113.9:4444
2026-09-26T18:40:02	NETWORK_CONNECTION	HIGH	10.0.0.5:49152 → 203.0.113.9:4444 (TCP)
2026-09-26T18:40:02	NETWORK_CONNECTION	INFO	10.0.0.5:51234 → 198.51.100.23:22 (TCP)
2026-09-26T18:40:02	NETWORK_CONNECTION	INFO	10.0.0.5:52000 → 10.0.0.7:445 (TCP)
2026-09-26T18:40:02	NETWORK_CONNECTION	INFO	10.0.0.5:53001 → 10.0.0.2:53 (UDP) dns=cdn-update.ngrok.io
2026-09-26T18:40:02	NETWORK_CONNECTION	HIGH	10.0.0.5:49152 → 203.0.113.9:4444 (TCP)
2026-09-26T18:40:02	NETWORK_CONNECTION	INFO	198.51.100.23:5911 → 10.0.0.5:22 (TCP)
2026-09-26T18:40:02	USER_ACCOUNT	INFO	User account svc_backup (last login 2026-09-26T10:02:10+00:00)
2026-09-26T18:40:02	USER_ACCOUNT	INFO	User account j.alice (last login 2026-09-26T10:00:05+00:00)
2026-09-26T18:40:02	USER_ACCOUNT	INFO	User account root (last login 2026-09-26T10:00:01+00:00)

Timestamp	Type	Severity	Description
2026-09-29T18:12:44	NETWORK_CONNECTIONNFO		198.51.100.23:5911 → 10.0.0.5:22 (TCP)
2026-09-29T18:16:10	NETWORK_CONNECTIONNFO		10.0.0.5:53001 → 10.0.0.2:53 (UDP) dns=cdn-update.ngrok.io
2026-09-29T18:16:20	NETWORK_CONNECTIONHIGH		10.0.0.5:49152 → 203.0.113.9:4444 (TCP)
2026-09-29T18:16:21	NETWORK_CONNECTIONHIGH		10.0.0.5:49152 → 203.0.113.9:4444 (TCP)

5. Chain of Custody / Audit

When	Actor	Action	Result
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Evidence hashes (first 40):

4326f0df... PROCESS sha256:12f9cd02ee00d6a5b30330f9289bfe3b7826619d61ffbbcaa462a97855dd2223
a71890be... PROCESS sha256:dfc0722b8cba2964963d7d67c0fe01f723c7585baf58b063d74a4b4a09fb5ac7
439024ca... PROCESS sha256:02902a4f644317fa8c2d38268cb1723a2652af57a256a18baa26e221ad5ce093
174bbf4b... PROCESS sha256:6ead899d1ef229277509f49404c6ad9658f333b2d74fe1fb1d239a0b8dbac4e7
16a587a7... FILE sha256:fccc3edfbd0c33875805bfc8dc8c65d02bf9d0d31adbf9dead97c643d1bf7116
f4cbf525... FILE sha256:d088528155c9ee1f506d2fddf81f08c871ae9caa2b3e97b21988f9bfa35dd3bc
fd3d1e2a... FILE sha256:036a5337bb9ea476357f95b71ee17f10115d42a71f95203beaaf8ca93ecaf06d
37dc5a94... NETWORK_CONNECTION
sha256:103a815dab2d6b855adb5fd29bffc9cd317c64ba964c57fa9a586997161a6f4d
c430d06e... NETWORK_CONNECTION
sha256:a74dc805a95b6377f759a3758accb2d8caa47578dd4637bdf5dd1a52b908d68a
bf94eb3a... NETWORK_CONNECTION
sha256:64a4dadfffd4a24be5c0acafade420b56e373abaae4167c16582859c2cab0743
44b41760... NETWORK_CONNECTION
sha256:c07b98919aa4e8242720a14f4484f1242851605ff64c86d33bafdd06b1cf5c5bd
6537944c... NETWORK_CONNECTION
sha256:21902a3aa86a0f23f147a494233fb6f6bf44fcbe5b884f62041e8d88d4dec0cf
9b1a239a... NETWORK_CONNECTION
sha256:6ed903cc36cc6568269506d54a5b2ccd338ba1ea46d117c8d4b001e13c39f702
adedb791... NETWORK_CONNECTION
sha256:9f69c9dd28663fc62e5bc160a6a92308c8656d3637b072f2be42c71a6a5aaa77f
cf664d5e... NETWORK_CONNECTION
sha256:95c332f66ddacb37dea20a720011a6fdefc1c01351ea87a421223c075541e9c6
02505bcb... NETWORK_CONNECTION
sha256:4e901917e40cf9d45746843c5f428edbe5deb89b6fe2ca1effea501beb1bbbae
4cd5caf7... NETWORK_CONNECTION
sha256:95338a805abcf97d91fc2a5bf0684aaeeb85226511c115c144165f71db94fba6
e6755622... USER_ACCOUNT sha256:6a3fc35ee7a1a97b717dd462441bc5aa88d8859fdc2cb0891b7ebab43ee2c2f3
461b4dde... USER_ACCOUNT sha256:f43735b0f573c3f638163cee6694e951ba4ac1d49d9079077affe81ab3972786
c0e8f8fb... USER_ACCOUNT sha256:cfe5ae97cf50a8fa082ae9b8471764dbb9cb057d51d52ff9ac020e46ea789035
62c19757... SYSTEM_EVENT sha256:24b436747482d4c9f93ccd088de6096e4a5e611a29f1bc9d577e22fda719ed2d
6950aceb... SYSTEM_EVENT sha256:61573a067b3c1d0ee4d0dd130309caf1557ecb17886d5251acf3ca3a50a30de8
c6756a7e... SYSTEM_EVENT sha256:913de2126159c8763dff7c50c7748a5d7d92a6a45b3b043046810ba09730109b
f2475860... SYSTEM_EVENT sha256:829713ba9d850bf2c24b7c6eb20c1b590a9d4019ed86946540daba8f47fad998
dcdaa736... SYSTEM_EVENT sha256:9734a2ced134089c849568868ce15aaffee7c98a3263d2385d5badf967864a30
999c018d... SYSTEM_EVENT sha256:1582d187fffaaf602debabcc19085e8780505ef4450acac5c1b22c402a832a3c
484abd57... SYSTEM_EVENT sha256:4ae3d8afae937896f0794b8063ab3037eb3601d144a95d0442df81cb5c30549c
7e660e00... SYSTEM_EVENT sha256:35b59001ed6a0ff0ab2e4d94ed96f0e9e171069887b0d18641a5626575aeddd85
5a195e54... SYSTEM_EVENT sha256:7b7d9b0006bfd9ed59680159e9b6922a0cd39f34ec49a7c9b2f52ae4819cd67a
905abcb4... SYSTEM_EVENT sha256:ea60e7e4a03ece79d6211d64d27244f4bb855f720403855c67f5464e17ffe097
508cdec9... SYSTEM_EVENT sha256:aeb56ade0b6f846ac55bbb3397d823102cab21ab800f51f84a97a8473643892
d0ad3443... SYSTEM_EVENT sha256:fccaa256b4dcde9cb71cfa93a028a6a631d848769c58e88d81e5090456f8f843
81af3669... SYSTEM_EVENT sha256:fb0c3ce46ad7ba7b05398c32ea61c2e48e22af6170212a4b41fb4d89d8915f14